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Friday, October 30, 2026 3:37 PM

3. $\frac{1}{1-x} = \sum_{j=0}^{\infty} x^j$ for $|x|<1$

Want to show

$\frac{1}{1-\theta z} = -\sum_{j=1}^{\infty} \theta^{-j} z^{-j}$ for $|\theta|>1$ and $|z| \geq 1$

Since $\theta z > 1$ by def'n, then $\left|\frac{1}{\theta z}\right| = |\theta^{-1} z^{-1}| < 1$

Now, we need to isolate $|\theta^{-1} z^{-1}|$ in $\frac{1}{1-\theta z}$

$$\frac{1}{1-\theta z} = \frac{1}{-\theta z\left(-\frac{1}{\theta z}+1\right)}$$

$$= \frac{1}{-\theta z\left(1-\frac{1}{\theta z}\right)}$$

$$= \frac{-1}{\theta z} \cdot \frac{1}{1-\frac{1}{\theta z}}$$

Let $x = \frac{1}{\theta z}$, knowing $|x| = \left|\frac{1}{\theta z}\right|$ both are < 1 satisfying the inequality.

$$\Rightarrow \frac{-1}{\theta z} \cdot \frac{1}{1-\frac{1}{\theta z}} = -x \cdot \frac{1}{1-x}$$

$$= \frac{-x}{1-x}$$

Now, we use the first geometric series involving x.

$$-x \cdot \frac{1}{1-x} = -x\left(\sum_{j=0}^{\infty} x^j\right)$$

$$= -\frac{1}{\theta z} \sum_{j=0}^{\infty} \left(\frac{1}{\theta z}\right)^j$$

$$= -(\theta^{-1} z^{-1}) \sum_{j=0}^{\infty} (\theta^{-1} z^{-1})^j$$

$$= -\sum_{j=0}^{\infty} (\theta^{-1} z^{-1})(\theta^{-j} z^{-j}) \quad \text{distribute into sum}$$

$$= -\sum_{j=0}^{\infty} (\theta^{-1-j} z^{-1-j})$$

$$= -\sum_{j=0}^{\infty} (\theta^{-(j+1)} z^{-(j+1)}) \quad \text{see that first index } j=0, \text{ first exponent} = -1$$

$$= -\sum_{j=1}^{\infty} (\theta^{-j} z^{-j}) \quad \text{thus, series is equivalent if first index } j=1, \text{ first exp.} = -1$$

∴ We have shown that $\frac{1}{1-\theta z} = \frac{-1}{\theta z} \cdot \frac{1}{1-(\theta^{-1} z^{-1})} = -x \cdot \frac{1}{1-x}$ is equal to $-\sum_{j=1}^{\infty} \theta^{-j} z^{-j}$.

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